7+1: provenance

Sources, derivations, corrections and the verification record of the notebook 7+1.nb and of its documentation

Repository: github.com/once-ere/7+1 — prepared for Nash
Notebook: 7+1.nb (Wolfram Language 15.0.1)
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Abstract

This record states, for every object of 7+1.nb, where it comes from (the task statement, the notebook 4+4.nb, or the literature), the equation that defines it, how it was derived, and how it was checked. It ends with the full list of the 128 checks and their results, the independent Python cross-checks, and the hashes of the files.

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1. Sources

1.1 The task statement

The task asked for a new Mathematica notebook 7+1.nb in `D:\Developer\github\7+1` defining a new realization of the octonions and a new representation of the Clifford algebra of an 8-dimensional vector space that mirror, as far as possible, the split octonions and the pseudo-Clifford algebra of 4+4.nb; real numbers and the real 8×8 and 16×16 matrices of 4+4.nb were to be used if possible, otherwise their complexification; examples were to check all work; notation was to be changed and documented where confusing. It fixed the coordinates

$$\begin{aligned}
&\text{4+4.nb:}\quad X=(x_0,\dots,x_7),\ x_0=\text{hidden space-like},\ x_1,x_2,x_3=\text{3-space},\\
&\qquad\qquad x_4=\text{time},\ x_5,x_6,x_7=\text{auxiliary time-like},\\
&\text{7+1.nb:}\quad X=(x_0,\dots,x_7),\ x_0=\text{time (octonion unit)},\ x_1,x_2,x_3=\text{3-space},\\
&\qquad\qquad x_4,\dots,x_7=\text{auxiliary space-like},
\end{aligned}$$

and gave two definitions. [1A] the octonions: $e_0=1$, $e_i^2=-1$, Fano plane, Cayley–Dickson doubling $\mathbb{O}=\mathbb{H}\oplus\mathbb{H}$, conjugation $\bar{x}=x_0-\sum_i x_i e_i$ with $\overline{xy}=\bar{y}\bar{x}$, $N(x)=x\bar{x}=\sum_A x_A^2$, $N(xy)=N(x)N(y)$, $x^{-1}=\bar{x}/N(x)$, alternativity, power-associativity. [1B] the Clifford algebra $Cl(V,Q)=\mathcal{T}(V)/\mathcal{I}_Q(V)$ with $v^2=Q(v)1$, $vw+wv=2B(v,w)1$, $B(v,w)=\frac{1}{2}(Q(v+w)-Q(v)-Q(w))$, dimension $2^8=256$, a classification table for $p+q=8$, and $Cl_8(\mathbb{C})\cong\mathbb{C}(16)$. Section 4 lists the statements of [1A] and [1B] that had to be corrected.

1.2 The notebook 4+4.nb

File `4+4.nb`, 9348656 bytes, written by Wolfram 15.0 (header `CreatedBy='Wolfram 15.0'`), SHA-256 `740023233eaa0cd24ca0e2e7010094671b1d314b698114c08db94050d3e0aed5`. It was read with the parser `tools/nbparse.py` (806 cells: 366 input, 366 output, 49 text). Every object used by 7+1.nb was reproduced with the formula of the cell given here, and each reproduction is checked in 7+1.nb section 2:

object	4+4.nb cell	definition in 4+4.nb
$\eta^{(4,4)}$	In[9]	$\mathrm{diag}(1_4,-1_4)$ (`η4488`)
σ	In[18]	$\left(\begin{smallmatrix}0 & 1_4\\ 1_4 & 0\end{smallmatrix}\right)$ (`σ=σ88`)
s_h, t_h	In[29], In[35], In[36]	$(s_h)_{pq}=\epsilon_{hpq4}-(\delta_{p4}\delta_{qh}-\delta_{ph}\delta_{q4})$, $(t_h)_{pq}=\epsilon_{hpq4}+(\dots)$
ζ_A	In[60]–In[63]	$\zeta_0=1_8$, $\zeta_{7-h}=\left(\begin{smallmatrix}0 & t_h\\ -t_h & 0\end{smallmatrix}\right)$, $\zeta_h=\left(\begin{smallmatrix}0 & s_h\\ s_h & 0\end{smallmatrix}\right)$, $\zeta_7=\zeta_1\cdots\zeta_6$
relations of ζ	In[64], In[65], In[67]	$\sigma=\zeta_1\zeta_2\zeta_3$; $\frac{1}{2}\{\zeta_A,\zeta_B\}=-\eta^{(4,4)}_{AB}1_8$; $-\eta^{(4,4)}_{AA}\zeta_A=\zeta_A^{\mathsf{T}}$
τ_A	In[70]	$\tau_0=1_8$, $\tau_A=\zeta_A$
$\bar{\tau}_A$	In[74]	$\bar{\tau}_0=1_8$, $\bar{\tau}_A=\sigma\tau_A^{\mathsf{T}}\sigma$ (symbolized tau-bar)
relations of τ	In[77], In[87], In[89]	$\frac{1}{2}(\tau_A\bar{\tau}_B+\tau_B\bar{\tau}_A)=\eta^{(4,4)}_{AB}1_8$; $\tau_1\cdots\tau_7=1_8$; $\sigma=\tau_4\tau_5\tau_6\tau_7$
T_A	In[98]	$T_A=\left(\begin{smallmatrix}0 & \bar{\tau}_A\\ \tau_A & 0\end{smallmatrix}\right)$ (`T16^A[A1]`)
T_8, σ_{16}	In[99], In[97], In[106]	$T_8=T_0\cdots T_7$; $\sigma_{16}=T_0T_1T_2T_3=\mathrm{diag}(-\sigma,\sigma)$
Clifford relation	In[110]	$\frac{1}{2}\{T_A,T_B\}=\eta^{(4,4)}_{AB}1_{16}$
P_L, P_R	In[112]–In[114]	$P_L=\frac{1}{2}(1_{16}-T_8)$, $P_R=\frac{1}{2}(1_{16}+T_8)$
U_h	In[257], In[259]	eigenvectors of σ divided by $\sqrt{2}$
u	In[264], In[266]	`hUSE = 8`, $u=U_8=(1,0,0,0,1,0,0,0)/\sqrt{2}$
g	In[283]	$g_{AB}=\eta^{(4,4)}$
E_{44}, E_{44}^{-1}	In[294]–In[296]	$(E_{44})_{Ba}=(U_h^{\mathsf{T}}\sigma\bar{\tau}_B)_a$; $E_{44}^{-1}=(U_h^{\mathsf{T}}\sigma U_h)(\eta^{(4,4)}_{BB}\tau_B U_h)_B$
ε_B	In[313]	`EA = Array[eA, 8]`
split table	In[325], Out[325]	$\varepsilon_B\varepsilon_C=\sum_D(g_{BB}E_{44}\tau_{B-1}E_{44}^{-1})_{DC}\varepsilon_D$ (`gridx`)

1.3 Literature

- J. C. Baez, *The Octonions*, Bull. Amer. Math. Soc. 39 (2002) 145–205, arXiv:math/0105155, section 2.2 (the page https://math.ucr.edu/home/baez/octonions/node5.html, which the task statement itself cites, was read on 2026-10-26): the Cayley–Dickson product $(a,b)(c,d) = (ac - db^*,\ a^*d + cb)$, where Baez writes x^* for the conjugate $\bar{x}$ and the conjugate of a pair is $(a,b)^* = (a^*, -b)$. 7+1.nb proves that its octonions satisfy this product with $x = a + \ell b$, $\ell = e_4$.
- H. B. Lawson and M.-L. Michelsohn, *Spin Geometry*, Princeton (1989): classification of real Clifford algebras, $Cl(p+1,q+1) \cong Cl(p,q) \otimes \mathbb{R}(2)$. 7+1.nb does not rely on it: it computes the classification directly (section 2.10 below).
- Standard facts used in proofs: Schur's lemma (the intertwiners of an irreducible module form a division algebra; over $\mathbb{C}$, the scalars); $Cl_8(\mathbb{C}) \cong \mathbb{C}(16)$ has a unique irreducible module $\mathbb{C}^{16}$; Hurwitz's theorem (a unital composition algebra of dimension 8 is the octonions if its norm is positive definite and the split octonions if the norm is indefinite); Degen's eight-square identity.

2. Derivations: how each new object was obtained

2.1 Octonion matrices τ^O

Origin: the split matrices of 4+4.nb and the observation (checked) that $\sigma = \tau_1\tau_2\tau_3 = \tau_4\tau_5\tau_6\tau_7$ commutes with τ_1, τ_2, τ_3, anticommutes with $\tau_4, \ldots, \tau_7$ and squares to 1_8. Hence

$$\tau_0^O = 1_8, \quad \tau_A^O = \tau_A \ (A \le 3), \quad \tau_k^O = \tau_k\sigma \ (k \ge 4), \qquad (\tau_k\sigma)^2 = \tau_k\sigma\tau_k\sigma = -\tau_k^2\sigma^2 = -1_8,$$
$$\tau_h^O\tau_k^O = \tau_h\tau_k\sigma = -\tau_k\tau_h\sigma = -\tau_k\sigma\tau_h = -\tau_k^O\tau_h^O, \qquad \tau_k^O\tau_l^O = \tau_k\sigma\tau_l\sigma = -\tau_k\tau_l = \tau_l\tau_k = -\tau_l^O\tau_k^O \quad (k \ne l \ge 4).$$

This is the real counterpart of the complex substitution $\tau_k \to i\tau_k$ that turns split octonions into octonions over $\mathbb{C}$.

2.2 Octonion vielbein and multiplication constants

Origin: In[294] and In[325] of 4+4.nb with the split spinor metric σ replaced by the Euclidean $g^O = 1_8$ and the same unit spinor u:

$$(E^O)_{Ba} = \left(u^{\mathsf{T}} g^O \bar{\tau}_B^O\right)_a, \qquad L_A^O = g_{AA}^O E^O \tau_A^O (E^O)^{-1}, \qquad e_A e_K = \sum_L (L_A^O)_{LK} e_L.$$

Why the result is the octonion algebra: $(E^O)^{-1}e_0 = u$ makes e_0 a two-sided unit, and $\tau^O(x)\bar{\tau}^O(x) = N(x)1_8$ together with the orthogonality of E^O gives $N(xy) = N(x)N(y)$. A unital composition algebra of dimension 8 with a positive definite norm is the octonions (Hurwitz). The notebook checks all the laws directly.

2.3 Cayley–Dickson form

Origin: Baez 2002, section 2.2. With $\ell = e_4$, $x = a + \ell b$, $a = (x_0, \ldots, x_3)$, $b = (x_4, \ldots, x_7)$:

$$(a,b)(c,d) = \left(ac + \gamma\, d\bar{b},\ \bar{a}d + cb\right), \qquad \gamma = -1 \text{ (7+1.nb)}, \qquad \gamma = +1 \text{ (4+4.nb)}.$$

2.4 Euclidean mirror T^O

Origin: In[98] of 4+4.nb with $\tau \to \tau^O$: $T_A^O = \begin{pmatrix} 0 & \bar{\tau}_A^O \\ \tau_A^O & 0 \end{pmatrix}$, so that $\{T_A^O, T_B^O\} = \begin{pmatrix} \bar{\tau}_A^O\tau_B^O + \bar{\tau}_B^O\tau_A^O & 0 \\ 0 & \tau_A^O\bar{\tau}_B^O + \tau_B^O\bar{\tau}_A^O \end{pmatrix} = 2\delta_{AB}1_{16}$. In terms of 4+4.nb: $T_A^O = T_A$ $(A \le 3)$, $T_k^O = T_k S_{44}$ $(k \ge 4)$, $S_{44} = T_4T_5T_6T_7 = \mathrm{diag}(\sigma, \sigma)$.

2.5 Real 7+1 generators $\Gamma^{(1,7)}$

Origin: the standard construction of a Lorentzian Clifford algebra from a Euclidean one, applied to T^O:

$$\Gamma_0^{(1,7)} = T_0^O, \qquad \Gamma_i^{(1,7)} = T_0^O T_i^O, \qquad (T_0^O T_i^O)^2 = -(T_0^O)^2 (T_i^O)^2 = -1_{16},$$
$$\{T_0^O, T_0^O T_i^O\} = T_i^O + T_0^O T_i^O T_0^O = T_i^O - T_i^O = 0, \qquad \{T_0^O T_i^O, T_0^O T_j^O\} = -\{T_i^O, T_j^O\} = -2\delta_{ij} 1_{16}.$$

Hence $\{\Gamma_A^{(1,7)}, \Gamma_B^{(1,7)}\} = 2\eta_{AB}^{(1,7)} 1_{16}$, and $\Gamma^{(1,7)}(x)^2 = \mathrm{Re}(xx) 1_{16}$.

2.6 Complexified generators $\Gamma^{(7,1)}$

Origin: the task's instruction to use the complexification of the 4+4 matrices if real numbers fail. Only the time block is multiplied by i: $\tau_0^C = \bar\tau_0^C = i 1_8$, $\tau_k^C = \tau_k^O$; then $\Gamma_A^{(7,1)} = \left(\begin{smallmatrix} 0 & \bar\tau_A^C \\ \tau_A^C & 0 \end{smallmatrix}\right)$, $(\Gamma_0^{(7,1)})^2 = i^2\, 1_{16} = -1_{16}$, and $\Gamma_A^{(7,1)} = S(i\Gamma_A^{(1,7)})S^{-1}$ with $S = (1_{16} - i\Gamma_0^{(1,7)})/\sqrt{2}$.

2.7 The impossibility theorem

Origin: Schur's lemma and the uniqueness of the irreducible module of $Cl_8(\mathbb{C}) \cong \mathbb{C}(16)$. The intertwiners M with $M\overline{\Gamma_A^{(7,1)}} = \Gamma_A^{(7,1)} M$ are computed in 7+1.nb with `NullSpace`; they form the line through $M_0 = \Gamma_8^{(1,7)} = \left(\begin{smallmatrix} 0 & -1_8 \\ 1_8 & 0 \end{smallmatrix}\right)$, and $M_0\overline{M_0} = -1_{16}$. A real realization $R_A = T\Gamma_A^{(7,1)}T^{-1}$ would give an intertwiner $M = T^{-1}\overline{T}$ with $M\overline{M} = 1_{16}$, impossible because $M = cM_0$ gives $M\overline{M} = -|c|^2 1_{16}$.

2.8 Real 32×32 generators

Origin: realification. $\widehat\Gamma_A^{(7,1)} = \varepsilon \otimes \Gamma_A^{(1,7)}$ with $\varepsilon^2 = -1_2$ gives $\{\widehat\Gamma_A^{(7,1)}, \widehat\Gamma_B^{(7,1)}\} = \varepsilon^2 \otimes 2\eta_{AB}^{(1,7)} 1_{16} = 2\eta_{AB}^{(7,1)} 1_{32}$. Commutant: $Q_i = \varepsilon \otimes 1_{16}$, $Q_j = \sigma_1 \otimes \Gamma_8^{(1,7)}$, $Q_k = \sigma_3 \otimes \Gamma_8^{(1,7)}$, found from $\varepsilon\sigma_1 = -\sigma_1\varepsilon$, $\varepsilon\sigma_3 = -\sigma_3\varepsilon$ and $\Gamma_8^{(1,7)}\Gamma_A^{(1,7)} = -\Gamma_A^{(1,7)}\Gamma_8^{(1,7)}$.

2.9 Examples

so(1,7): $\Sigma_{AB} = \frac{1}{4}[\Gamma_A^{(1,7)}, \Gamma_B^{(1,7)}]$; the commutation relations follow from the Clifford relation. Boost and rotation: $(\Gamma_0^{(1,7)}\Gamma_1^{(1,7)})^2 = 1_{16}$, $(\Gamma_4^{(1,7)}\Gamma_5^{(1,7)})^2 = -1_{16}$, so $e^{\beta\Sigma_{01}} = \cosh\frac{\beta}{2} + \sinh\frac{\beta}{2}\Gamma_0^{(1,7)}\Gamma_1^{(1,7)}$ and $e^{\theta\Sigma_{45}} = \cos\frac{\theta}{2} + \sin\frac{\theta}{2}\Gamma_4^{(1,7)}\Gamma_5^{(1,7)}$. Dirac operator: $(\Gamma^A\partial_A)^2 = \frac{1}{2}\{\Gamma^A, \Gamma^B\}\partial_A\partial_B = \eta_{(1,7)}^{AB}\partial_A\partial_B$. Real amplitudes: $p = (3,2,2,0,0,0,0,0)$, $p^{\mathsf T}\eta^{(1,7)}p = 1$, $\not{p}^{\,2} = 1_{16}$; the real matrix $\not{p}$ has a real 8-dimensional eigenspace for the eigenvalue $+1$.

Real spinor fields. If J anticommutes with every $\Gamma_A^{(1,7)}$, then $J\Gamma_8^{(1,7)}$ commutes with every $\Gamma_A^{(1,7)}$ (because $\Gamma_8^{(1,7)}$ anticommutes with each of them), so by irreducibility $J\Gamma_8^{(1,7)} = c\, 1_{16}$ and $J = -c\,\Gamma_8^{(1,7)}$ (as $(\Gamma_8^{(1,7)})^{-1} = -\Gamma_8^{(1,7)}$), whose square is $-c^2\, 1_{16}$. Hence the real operator $\Gamma^A\partial_A - mJ$ has the square $\partial_0^2 - \cdots - \partial_7^2 - c^2m^2$ (a tachyon), and a real equation $(G^A\partial_A - m)\Psi = 0$ implies the Klein–Gordon equation only if the real G_A obey the $\eta^{(7,1)}$ relation, impossible for 16×16 matrices (2.7). Massless: for $k = (1,1,0,\ldots,0)$, $\not{k}^{\,2} = k^{\mathsf T}\eta^{(1,7)}k\, 1_{16} = 0$ and $\Gamma^A\partial_A[w\cos(k\cdot x)] = -\sin(k\cdot x)\ \not{k}\, w$, which vanishes for the 8 real solutions of $\not{k}\, w = 0$. Massive, with the real 32×32 matrices: $P = \sum_A p^A\widehat\Gamma_A^{(7,1)}$, $P^2 = -m^2\, 1_{32}$, and with $\phi = p\cdot x$, $\widehat\Gamma^A = \eta_{(7,1)}^{AA}\widehat\Gamma_A^{(7,1)}$:

$$\begin{aligned}\widehat\Gamma^A\partial_A\Big(w\cos\phi + \tfrac{1}{m}Pw\sin\phi\Big) &= -P\Big(-w\sin\phi + \tfrac{1}{m}Pw\cos\phi\Big)\\ &= Pw\sin\phi + m\,w\cos\phi = m\Big(w\cos\phi + \tfrac{1}{m}Pw\sin\phi\Big),\end{aligned}$$

using $\partial_A\phi = (\eta^{(1,7)}p)_A$ and $\eta_{(7,1)}^{AA}\eta_{AA}^{(1,7)} = -1$.

2.10 Classification of the nine signatures

For each p, complex generators $G_A = T_A^O$ $(A < p)$, $G_A = iT_A^O$ $(A \geq p)$ satisfy $\{G_A, G_B\} = \pm 2\delta_{AB} 1_{16}$ with p plus signs. The indicator sign of $M\overline{M}$ ($M\overline{G_A} = G_A M$) is $+1$ for $p \in \{8,5,4,1,0\}$ ($\mathbb{R}(16)$) and -1 for $p \in \{7,6,3,2\}$ ($\mathbb{H}(8)$). An independent numerical method (`tools/classify_clifford.py`: commutant of the realified 32-dimensional representation, whose traceless part has a negative definite trace form exactly in the quaternionic case) gives the same table. Explicit real generators: T^O (8,0), $\Gamma_0^{(5,3)} = T_0$, $\Gamma_i^{(5,3)} = T_0 T_i$ (5,3) with $\eta^{(5,3)} = \mathrm{diag}(1,-1,-1,-1,1,1,1,1)$, T (4,4), $\Gamma^{(1,7)}$ (1,7), $\Gamma_A^{(0,8)} = T_8^O T_A^O$ (0,8).

3. Corrections to the task statement (with the evidence in 7+1.nb)

statement in the task	correction	evidence (check in 7+1.nb)
[1A]: $(a,b)(c,d) = (ac - \bar{d}b,\ \bar{a}d + cb)$ defines the octonions	not alternative, norm not multiplicative; correct form $(ac - d\bar{b},\ \bar{a}d + cb)$ (Baez 2002, 2.2)	"the formula ... of [1A] is NOT alternative", "... its norm is not multiplicative", "Cayley-Dickson, γ = -1 ..."
[1A]: first formula $(ac - \bar{d}b,\ da + cb\bar{x})$	contains a stray "x"; not interpretable	—
[1B]: 1+8+28+56+72+56+28+8+1 = 256	$\binom{8}{4} = 70$	"grades: binomial(8,k) == ..."
[1B]: $Cl(7,1) \cong \mathbb{R}(16)$	$Cl(7,1) \cong \mathbb{H}(8)$ for $v^2 = Q(v)$	"indicator(Γ71) == -1 ...", "classification: ..."
[1B]: $Cl(5,3) \cong \mathbb{C}(16)$, $Cl(3,5) \cong \mathbb{C}(16)$	$Cl(5,3) \cong \mathbb{R}(16)$, $Cl(3,5) \cong \mathbb{H}(8)$; $\mathbb{C}(16)$ has real dimension 512	"classification: ...", "$\mathbb{C}(16)$ cannot occur ..."
[1B]: metric-sign strings for (6,2), (5,3), (4,4), (2,6)	mistyped (wrong number of signs)	table of section 5.6 of 7+1.nb
split octonions are a "pseudo-normed division algebra"	they have zero divisors: $(\varepsilon_1 + \varepsilon_5)(\varepsilon_1 - \varepsilon_5) = 0$	"split: (eA[1] + eA[5])(eA[1] - eA[5]) == 0 ..."

4. Verification record

4.1 The Mathematica run

item	value
program	`C:\Program Files\Wolfram Research\Wolfram\15.0.1\WolframNB.exe`
$Version	15.0.1 for Microsoft Windows (64-bit) (July 2, 2026)
method	`Get["D:/Developer/github/7+1/tools/run_eval.wl"]`: `NotebookEvaluate[nb, InsertResults -> True]`, then `NotebookSave`
finished	Mon 26 Oct 2026 14:54:43
evaluation time	5.09545 s
private context of the notebook	`Notebook$$43$40478`
input cells / output cells / message cells	56 / 154 / 0
checks / True	128 / 128
commutant computation (8192×1024 sparse `NullSpace`)	1.0349715 s
SHA-256 of the evaluated 7+1.nb (306963 bytes)	`f7abe7279b5540e914de876dc055c199bd491b9c2d9522bf28dbc23689e237ce`

4.2 All checks of 7+1.nb and their results

Each row is one cell `check["name", test]` of the notebook, in order, grouped by the section of the notebook in which it stands; the test is the Wolfram Language expression in that cell, and the result is what Wolfram 15.0.1 returned.

2.1 σ, the self-dual and anti-self-dual blocks, ζ, τ and τbar

#	check	result
1	4+4: ζ[7] == diag(-1,-1,-1,-1,1,1,1,1)	True
2	4+4: σ == ζ[1].ζ[2].ζ[3]	True
3	4+4: (ζ[A].ζ[B] + ζ[B].ζ[A])/2 == -η4488[[A+1,B+1]] ID8 (A,B = 1..7)	True
4	4+4: -η4488[[A+1,A+1]] ζ[A] == Transpose[ζ[A]] (A = 1..7)	True
5	4+4: (τ[A].τbar[B] + τ[B].τbar[A])/2 == η4488[[A+1,B+1]] ID8	True
6	4+4: (τbar[A].τ[B] + τbar[B].τ[A])/2 == η4488[[A+1,B+1]] ID8	True
7	4+4: τ[1].τ[2]...τ[7] == ID8	True
8	4+4: σ == τ[1].τ[2].τ[3] == τ[4].τ[5].τ[6].τ[7]	True
9	4+4: σ commutes with τ[1..3] and anticommutes with τ[4..7]; σ.σ == ID8	True

2.2 The 4+4 Dirac matrices T16 and the chiral projectors

#	check	result
10	4+4: (T16[A].T16[B] + T16[B].T16[A])/2 == η4488[[A+1,B+1]] ID16	True
11	4+4: T16[A] real (entries 0, ±1)	True
12	4+4: T16[8] == diag(-ID8, ID8)	True
13	4+4: σ16 == diag(-σ, σ)	True
14	4+4: PL44 + PR44 == ID16, PL44.PL44 == PL44, PR44.PR44 == PR44, PL44.PR44 == 0	True

2.3 The split-octonion multiplication constants (gridx of 4+4.nb) and how they come from τ

#	check	result
15	4+4: uBasis == evecs/Sqrt[2] from Eigensystem[σ] (the basis of 4+4.nb)	True
16	4+4: unit == (1,0,0,0,1,0,0,0)/Sqrt[2], σ.unit == unit, unit.unit == 1, unit.σ.unit == 1	True
17	4+4 (In[294]): E44inv[h] == (u.σ.u) Transpose[Table[η4488[[B,B]] τ[B-1].u, {B,1,8}]] for h = 1..8	True
18	4+4: gridx44 is identical to the split-octonion table printed by 4+4.nb	True
19	4+4: LS[A] are signed permutation matrices (constants 0, ±1)	True

2.4 Split octonions: composition algebra, but not a division algebra

#	check	result
20	split: eA[1] is the identity; eA[2..4]^2 == -eA[1]; eA[5..8]^2 == +eA[1]	True
21	split: x x̄ == N(x) eA[1] with N = η4488 (symbolic)	True
22	split: N(xy) == N(x) N(y) (symbolic)	True
23	split: alternative x(xy) == (xx)y and (yx)x == y(xx) (symbolic)	True
24	split: (eA[1] + eA[5])(eA[1] - eA[5]) == 0 with both factors nonzero: NOT a division algebra	True

3.1 From the split matrices τ to the octonionic matrices τO

#	check	result
25	τO[A] (A = 1..7): real, antisymmetric, τO[A].τO[A] == -ID8	True
26	τO[A] anticommute: τO[A].τO[B] == -τO[B].τO[A] (A != B, A,B = 1..7)	True
27	mirror of 4+4: (τO[A].τObar[B] + τO[B].τObar[A])/2 == gO[[A+1,B+1]] ID8	True
28	mirror of 4+4: (τObar[A].τO[B] + τObar[B].τO[A])/2 == gO[[A+1,B+1]] ID8	True

#	check	result
29	mirror of 4+4: τO[1].τO[2]...τO[7] == ID8	True
30	mirror of 4+4: τObar[A] == gO.Transpose[τO[A]].gO, as τbar[A] == σ.Transpose[τ[A]].σ	True
31	τO[4..7] are products of three split τ: τO[4] == τ[5].τ[6].τ[7], τO[5] == -τ[4].τ[6].τ[7], τO[6] == τ[4].τ[5].τ[7], τO[7] == -τ[4].τ[5].τ[6]	True

3.2 The octonion vielbein (the same unit spinor u as in 4+4.nb)

#	check	result
32	σ.unit == unit, hence τO[A].unit == τ[A].unit (A = 0..7)	True
33	EO is orthogonal: EOinv == Transpose[EO]	True
34	mirror of 4+4.nb In[294]: EOinv == (u.gO.u) Transpose[Table[gO[[B,B]] τO[B-1].u, {B,1,8}]]	True
35	EO == η4488.E44[hUSE]: the two vielbeins differ only by the signs of rows 5..8	True

3.3 The octonion multiplication constants: gridO, the mirror of gridx

#	check	result
36	gridO equals the octonion table computed independently in Python	True
37	LO[A] are signed permutation matrices (all multiplication constants are 0, ±1)	True
38	oMul reproduces gridO on all 64 pairs of basis units	True

3.4 The properties listed in [1A], verified

#	check	result
39	e[0] is a two-sided identity	True
40	e[i] e[i] == -e[0] (i = 1..7)	True
41	e[i] e[j] == -e[j] e[i] (i != j, i,j = 1..7): non-commutative	True
42	7 Fano lines; every pair of distinct imaginary units lies on exactly one line	True
43	each oriented line (a,b,c) is cyclic: e[a]e[b] == e[c], e[b]e[c] == e[a], e[c]e[a] == e[b]	True
44	each line spans a quaternion subalgebra: {e[0], e[a], e[b], e[c]} closed and associative	True
45	the picture's 7 lines are exactly fanoLines	True
46	left alternative: x(xy) == (xx)y (symbolic)	True
47	right alternative: (yx)x == y(xx) (symbolic)	True
48	flexible: (xy)x == x(yx) (symbolic)	True
49	associator is alternating: [x,y,z] == -[y,x,z] == -[x,z,y] (symbolic)	True
50	power associativity: x(xx) == (xx)x (symbolic)	True
51	Moufang: z(x(zy)) == ((zx)z)y, x(z(yz)) == ((xz)y)z, (zx)(yz) == (z(xy))z (x, y basis units, z symbolic)	True
52	x x̄ == x̄ x == N(x) e[0] (symbolic)	True
53	conjugation is an anti-automorphism: conj(xy) == conj(y) conj(x) (symbolic)	True
54	Re(x) == (x + x̄)/2 == x0	True
55	norm is multiplicative: N(xy) == N(x) N(y) (symbolic; Degen's eight-square identity)	True
56	inverse x^-1 = x̄/N(x): x x^-1 == x^-1 x == e[0] (symbolic)	True
57	every x satisfies x x - 2 Re(x) x + N(x) e[0] == 0, so the subalgebra generated by x is ℝ or ℂ	True
58	exactly 168 of the 512 ordered basis triples are non-associative (= 7*6*4: distinct imaginary units not on one Fano line)	True
59	example: (e[1]e[2])e[4] == e[7] but e[1](e[2]e[4]) == -e[7]	True
60	numerical example: N(x y) == N(x) N(y) == 15*6 == 90 and x x^-1 == e[0]	True
61	numerical example: x y != y x and (x y) z != x (y z)	True

3.5 Cayley–Dickson doubling: $\mathbb{O} = \mathbb{H} \oplus \ell\mathbb{H}$

#	check	result
62	$\mathbb{H}$ = span{e[0],e[1],e[2],e[3]} is a subalgebra	True
63	ℓ = e[4] doubles $\mathbb{H}$: e[4] e[k] == e[4+k] (k = 1,2,3) and e[4] e[4] == -e[0]	True
64	orientation inherited from 4+4.nb: e[1] e[2] == -e[3]	True
65	Cayley-Dickson, γ = -1: (ac - d $\bar{b}$, $\bar{a}$ d + c b) reproduces the octonion product (symbolic)	True
66	Cayley-Dickson, γ = +1: (ac + d $\bar{b}$, $\bar{a}$ d + c b) reproduces the SPLIT product of 4+4.nb (symbolic)	True
67	the formula (ac - $\bar{d}$ b, $\bar{a}$ d + c b) of [1A] is NOT alternative: x·x·y != x(xy) for x = e[0]+e[4], y = e[1]	True
68	... and its norm is not multiplicative (symbolic)	True

3.6 Octonions versus split octonions

#	check	result
69	e[A]e[B] (octonion) == s_AB e[A]e[B] (split) for all A, B = 0..7	True
70	over $\mathbb{C}$: ϕ(x *split y) == ϕ(x) *oct ϕ(y) (symbolic): split $\otimes\ \mathbb{C} \cong \mathbb{O} \otimes \mathbb{C}$	True
71	the norms correspond: N_oct(ϕ(x)) == N_split(x) (symbolic)	True

4.1 The exact mirror of T16: the Euclidean Clifford algebra Cl(8,0)

#	check	result
72	Cl(8,0): (T16O[A].T16O[B] + T16O[B].T16O[A])/2 == gO[[A+1,B+1]] ID16	True
73	S44 == diag(σ, σ)	True
74	T16O from the 4+4 matrices: T16O[A] == T16[A] (A = 0..3), T16O[k] == T16[k].S44 (k = 4..7)	True
75	T16O[8] == T16[8] == diag(-ID8, ID8): same chirality matrix and same PL44, PR44 as 4+4.nb	True

4.2 The REAL Clifford algebra of 7+1 spacetime: Γ17, Cl(1,7) ≅ $\mathbb{R}$(16)

#	check	result
76	7+1 REAL Clifford relation: (Γ17[A].Γ17[B] + Γ17[B].Γ17[A])/2 == η17[[A+1,B+1]] ID16 == -η71[[A+1,B+1]] ID16	True
77	Γ17[A] are REAL: every entry is 0 or ±1	True
78	block form: Γ17[0] == {{0,ID8},{ID8,0}}, Γ17[i] == {{τO[i],0},{0,τObar[i]}}	True
79	built from the real 4+4 matrices: Γ17[0] == T16[0], Γ17[i] == T16[0].T16[i] (i = 1,2,3), Γ17[k] == T16[0].T16[k].S44 (k = 4..7)	True
80	Γ17[0] symmetric, Γ17[i] antisymmetric, so Transpose[Γ17[A]] == Γ17[0].Γ17[A].Γ17[0] (real Dirac adjoint)	True
81	the 256 products Γ17[A1]...Γ17[Ak] (A1 < ... < Ak) are linearly independent: Cl(1,7) ≅ $\mathbb{R}$(16)	True
82	grades: binomial(8,k) == {1,8,28,56,70,56,28,8,1}, total 256 (binomial(8,4) == 70, not 72)	True
83	Tr[product] == 0 for every non-empty product, so Tr[M]/16 is the scalar part of M	True
84	irreducible over $\mathbb{R}$: only multiples of ID16 commute with all Γ17[A] (Schur; real commutant of dimension 1)	True
85	universal property: Γ17(v).Γ17(v) == (v.η17.v) ID16 for a symbolic vector v = X	True

4.3 The complexified 7+1 realization in the exact block form of 4+4.nb: Γ71

#	check	result
86	7+1 COMPLEX Clifford relation, 4+4.nb sign: (Γ71[A].Γ71[B] + Γ71[B].Γ71[A])/2 == η71[[A+1,B+1]] ID16	True
87	Γ71[0] == I T16[0] and Γ71[k] == T16O[k] (k = 1..7): only the time component is complex	True
88	S = (ID16 - I Γ17[0])/Sqrt[2] is unitary and Γ71[A] == S.(I Γ17[A]).Inverse[S]	True

4.4 Theorem: with η71 there is no real 16×16 realization, because Cl(7,1) ≅ $\mathbb{H}$(8)

#	check	result
89	Schur: the intertwiners M with M.Conjugate[Γ71[A]] == Γ71[A].M form a 1-dimensional space	True
90	... spanned by the volume element Γ17[8] == {{0,-ID8},{ID8,0}}, and Γ17[8].Γ17[8] == -ID16	True
91	indicator(Γ71) == -1: Cl(7,1) ≅ ℍ(8), NO real 16×16 realization of the η71 relation	True
92	indicator(Γ17) == +1: Cl(1,7) ≅ ℝ(16), real realization exists (Γ17)	True

4.5 A real realization of Cl(7,1) needs 32×32 matrices

#	check	result
93	REAL 32×32 realization of the η71 relation: (Γ71R32[A].Γ71R32[B] + Γ71R32[B].Γ71R32[A])/2 == η71[[A+1,B+1]] ID32	True
94	Qi, Qj, Qk commute with every Γ71R32[A]; Qi.Qi == Qj.Qj == Qk.Qk == -ID32; Qi.Qj == Qk == -Qj.Qi	True
95	the commutant of {Γ71R32[A]} has dimension exactly 4 and is spanned by ID32, Qi, Qj, Qk: it is ℍ	True

4.6 Volume elements and chirality

#	check	result
96	Γ17[8].Γ17[8] == -ID16 and Γ17[8] anticommutes with every Γ17[A]: no real chirality in 7+1	True
97	Γ71[8].Γ71[8] == -ID16 and χ71 = -I Γ71[8] == T16O[8] == T16[8] (real, diagonal)	True
98	7+1 chiral projectors in the Γ71 basis are exactly PL44, PR44 of 4+4.nb; χ71 anticommutes with every Γ71[A]	True
99	complex chiral projectors for Γ17: PL17 + PR17 == ID16, PL17.PL17 == PL17, PL17.PR17 == 0, rank 8 each	True

4.7 What the octonions do: Γ(x)² = Re(x x), and alternativity

#	check	result
100	Re(x x) == X.η17.X: the octonion square produces the 7+1 interval (symbolic)	True
101	octonion basis: Γ17e[0] == {{0,ID8},{ID8,0}} and Γ17e[i] == {{LO[i],0},{0,-LO[i]}}	True
102	alternativity in action: xImag(xImag ψ) == -|xImag|^2 ψ for the imaginary part xImag of x (symbolic)	True
103	T16O(x)^2 == N(x) ID16 (octonion norm) and T16(x)^2 == (X.η4488.X) ID16 (4+4.nb)	True

5.1 The Lorentz algebra so(1,7)

#	check	result
104	28 real generators Σ17[A,B] (A < B), linearly independent	True
105	[Σ17[A,B], Γ17[K]] == η17[[B+1,K+1]] Γ17[A] - η17[[A+1,K+1]] Γ17[B]	True
106	so(1,7): [Σ_AB, Σ_KL] == η_BK Σ_AL - η_AK Σ_BL - η_BL Σ_AK + η_AL Σ_BK	True
107	the complex structure J = Γ17[8] commutes with all Σ17[A,B] (Spin(1,7) acts complex-linearly on ℝ^16 = ℂ^8)	True

5.2 A boost along x1 and a rotation in the x4–x5 plane

#	check	result
108	S(β) and R(θ) are the exponentials: dS/dβ == Σ17[0,1].S, S(0) == ID16, dR/dθ == Σ17[4,5].R, R(0) == ID16, and S.Sinv == R.Rinv == ID16	True
109	S(β).Γ17[K].S(β)^-1 == Σ_B Λboost[[B+1,K+1]] Γ17[B]; Λboost is the boost along x1	True
110	R(θ).Γ17[K].R(θ)^-1 == Σ_B Λrot[[B+1,K+1]] Γ17[B]; Λrot is the rotation in the x4-x5 plane	True
111	both Λ preserve the 7+1 metric: Transpose[Λ].η17.Λ == η17	True

5.3 The 7+1 Dirac operator squares to the 7+1 wave operator

#	check	result
112	(Γ17^A ∂_A)^2 Ψ == (∂0^2 - ∂1^2 - ... - ∂7^2) Ψ for 16 arbitrary functions of x0..x7	True

5.4 Spinors on the mass shell: real amplitudes

#	check	result
113	p is on the mass shell (p.η17.p == m^2) and pslash.pslash == m^2 ID16	True
114	(pslash - m) w == 0 has 8 linearly independent REAL solutions w	True
115	Ψ = w Exp[-I p.x] solves (I Γ17^A ∂_A - m) Ψ == 0	True

5.5 Real spinor fields: massless with Γ17, massive with the real 32×32 matrices

#	check	result
116	the matrices anticommuting with every Γ17[A] are exactly the multiples of Γ17[8], and (c Γ17[8])^2 == -c^2 ID16: no real J with J.J == +ID16	True
117	null momentum k = (1,1,0,0,0,0,0,0): k.η17.k == 0, kslash.kslash == 0, and kslash.w == 0 has 8 REAL solutions	True
118	Ψ = w Cos[k.x] is a REAL solution of the real massless Dirac equation Γ17^A ∂_A Ψ == 0	True
119	(Γ71R32^A ∂_A)^2 Ψ == -(∂0^2 - ∂1^2 - ... - ∂7^2) Ψ for 32 arbitrary functions of x0..x7	True
120	p.η71.p == -m^2 and pslash32.pslash32 == -m^2 ID32	True
121	Ψ = w Cos[p.x] + (pslash32.w/m) Sin[p.x], with 32 arbitrary REAL constants w, solves the REAL massive equation (Γ71R32^A ∂_A - m) Ψ == 0	True
122	... and Ψ obeys the Klein-Gordon equation (∂0^2 - ∂1^2 - ... - ∂7^2 + m^2) Ψ == 0	True

5.6 All nine signatures p + q = 8, and the corrected table of [1B]

#	check	result
123	split analogue: Re(x x) of split octonions == X.η53.X, and Γ53 realizes Cl(5,3) with REAL 16×16 matrices	True
124	Γ08[A] = T16O[8].T16O[A] realizes Cl(0,8) with REAL 16×16 matrices	True
125	gens[p] satisfy the Clifford relation of signature (p, 8-p) for p = 0..8	True
126	classification: ℝ(16) for (8,0),(5,3),(4,4),(1,7),(0,8); ℍ(8) for (7,1),(6,2),(3,5),(2,6)	True
127	every ℝ(16) case has explicit REAL 16×16 generators in this notebook (T16O, Γ53, T16, Γ17, Γ08)	True
128	ℂ(16) cannot occur: its real dimension 2*16^2 = 512 differs from 2^8 = 256	True

4.3 Independent Python cross-checks

tools/verify_7p1_nb.py (parses the evaluated 7+1.nb; exit code 0)

```
input cells                                                          : 56
output cells                                                         : 154
message cells (must be 0)                                            : 0
check outputs                                                        : 128
check outputs that are True                                          : 128
check["..."] calls in the input cells                                : 128
$Version output                                                      : ['{"15.0.1 for
    ...Microsoft Windows (64-bit) (July 2, 2026)","Notebook$$43$40478`"}']
private notebook context                                             : True
Length[checks]                                                       : 128
failedChecks == {}                                                   : True
allChecksPassed output True                                          : True
tau[A] printed == 4+4 tau[A] (python)                                : True
tauO[A] printed == python tauO[A]                                    : True
Gamma17[A] printed == python G17[A]                                  : True
gridO printed == python octonion table                               : True
gridx44 printed == python split table (4+4.nb)                       : True
Lambda boost/rotation printed                                        : True
8 real on-shell spinors printed solve (pslash - m) w = 0 (python check) : True
classification table printed                                         : True

VERDICT: ALL CROSS-CHECKS PASSED
```

tools/ref44.py (rebuilds 4+4.nb; exit code 0)

```
ζ7 = [-1, -1, -1, -1, 1, 1, 1, 1]
σ == ζ1ζ2ζ3 : True
{ζA,ζB}/2 == -η ID8 (A,B=1..7): True
-η_AA ζA == ζA^T : True
(τA τB+τB τA)/2 == η ID8 : True
(τA̅ τB+τB τA)/2 == η ID8 : True
τ1..τ7 product == ID8: True
σ == τ4τ5τ6τ7: True
{T16A,T16B}/2 == η ID16 : True
T16[8] diag: [-1, -1, -1, -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1]
σ16 == diag(-σ,σ): True
unit: [sqrt(2)/2, 0, 0, 0, sqrt(2)/2, 0, 0, 0]  unit.σ.unit = 1
E^a_A[h] == (u σ u) Transpose[Table[η_BB τ[B-1].u]] for all h: True
eA[1] : ['eA[1]', 'eA[2]', 'eA[3]', 'eA[4]', 'eA[5]', 'eA[6]', 'eA[7]', 'eA[8]']
eA[2] : ['eA[2]', '-eA[1]', '-eA[4]', 'eA[3]', '-eA[6]', 'eA[5]', 'eA[8]', '-eA[7]']
eA[3] : ['eA[3]', 'eA[4]', '-eA[1]', '-eA[2]', '-eA[7]', '-eA[8]', 'eA[5]', 'eA[6]']
eA[4] : ['eA[4]', '-eA[3]', 'eA[2]', '-eA[1]', '-eA[8]', 'eA[7]', '-eA[6]', 'eA[5]']
```

```
eA[5] : ['eA[5]', 'eA[6]', 'eA[7]', 'eA[8]', 'eA[1]', 'eA[2]', 'eA[3]', 'eA[4]']
eA[6] : ['eA[6]', '-eA[5]', 'eA[8]', '-eA[7]', '-eA[2]', 'eA[1]', '-eA[4]', 'eA[3]']
eA[7] : ['eA[7]', '-eA[8]', '-eA[5]', 'eA[6]', '-eA[3]', 'eA[4]', 'eA[1]', '-eA[2]']
eA[8] : ['eA[8]', 'eA[7]', '-eA[6]', '-eA[5]', '-eA[4]', '-eA[3]', 'eA[2]', 'eA[1]']
```

tools/ref71.py (rebuilds 7+1.nb objects; exit code 0)

```
tauO antisymmetric (A>=1)                          : True
tauO[A]^2=-1 (A>=1)                                : True
tauO Euclid relation                               : True
tauObar tauO relation                              : True
tauO1..7 product                                   : [1, 1, 1, 1, 1, 1, 1, 1]
sigma commutes tau1..3, anticommutes tau4..7       : True
E orthogonal                                       : True
oct: unit two-sided                                : True
oct: e_i^2                                         : [-1, -1, -1, -1, -1, -1, -1]
oct: x xbar = N(x) 1                               : True
oct: N(xy)=N(x)N(y)                                : True
oct: conj(xy)=conj(y)conj(x)                       : True
oct: alternative (left,right,flexible)             : True
oct: Moufang identities                            : True
oct: #nonassociative basis triples                 : 168
oct: #noncommuting ordered basis pairs             : 42
e_A*e_B (O) = s_AB e_A*e_B (S)                     : True
T16O Euclid Cl(8,0)                                : True
T16O[A]==T16A44[A] (A<=3)                          : True
T16O[k]==T16A44[k].S44 (k>=4)                      : True
S44==diag(sigma,sigma)                             : True
T16O[8] diag                                       : [-1, -1, -1, -1, -1, -1, -1, -1, 1, 1, 1,
    ...1, 1, 1, 1, 1]
G17 Cl(1,7)                                        : True
G17 real                                           : True
G17[0] sym, G17[i] antisym                         : True
G17[8]^2                                           : -1
G17 blocks: G17[i]==diag(tauO,tauObar)             : True
G71 Cl(7,1) eta71                                  : True
G71[8]^2                                           : -1
G71 = Ssim (I G17) Ssim^-1                         : True
G71R32 Cl(7,1) real                                : True
```

tools/extra71.py (further identities; exit code 0)

```
Lambda boost (rows B = 0..7):
    ['cosh(beta)', '-sinh(beta)', '0', '0', '0', '0', '0', '0']
    ['-sinh(beta)', 'cosh(beta)', '0', '0', '0', '0', '0', '0']
    ['0', '0', '1', '0', '0', '0', '0', '0']
    ['0', '0', '0', '1', '0', '0', '0', '0']
    ['0', '0', '0', '0', '1', '0', '0', '0']
    ['0', '0', '0', '0', '0', '1', '0', '0']
    ['0', '0', '0', '0', '0', '0', '1', '0']
    ['0', '0', '0', '0', '0', '0', '0', '1']
Lambda rotation (rows B = 0..7):
    ['1', '0', '0', '0', '0', '0', '0', '0']
    ['0', '1', '0', '0', '0', '0', '0', '0']
    ['0', '0', '1', '0', '0', '0', '0', '0']
    ['0', '0', '0', '1', '0', '0', '0', '0']
    ['0', '0', '0', '0', 'cos(theta)', '-sin(theta)', '0', '0']
    ['0', '0', '0', '0', 'sin(theta)', 'cos(theta)', '0', '0']
    ['0', '0', '0', '0', '0', '0', '1', '0']
    ['0', '0', '0', '0', '0', '0', '0', '1']
indicator classification (signature, number of intertwiners, M.conj(M) for the normalized M,
```

```
    ...algebra):
    Cl(8,0): 1 intertwiner, M.conj(M) = +0.0625 * 1  ->  R(16)
    Cl(7,1): 1 intertwiner, M.conj(M) = -0.0625 * 1  ->  H(8)
    Cl(6,2): 1 intertwiner, M.conj(M) = -0.0625 * 1  ->  H(8)
    Cl(5,3): 1 intertwiner, M.conj(M) = +0.0625 * 1  ->  R(16)
    Cl(4,4): 1 intertwiner, M.conj(M) = +0.0625 * 1  ->  R(16)
    Cl(3,5): 1 intertwiner, M.conj(M) = -0.0625 * 1  ->  H(8)
    Cl(2,6): 1 intertwiner, M.conj(M) = -0.0625 * 1  ->  H(8)
    Cl(1,7): 1 intertwiner, M.conj(M) = +0.0625 * 1  ->  R(16)
    Cl(0,8): 1 intertwiner, M.conj(M) = +0.0625 * 1  ->  R(16)
EO == eta4488 . E44[8]                                              : True
G53 Cl(5,3) real                                                    : True
G08 Cl(0,8) real                                                    : True
so(1,7): [S_AB,S_CD]=eta_BC S_AD - eta_AC S_BD - eta_BD S_AC + eta_AD S_BC: True
[S_AB,G_C]=eta_BC G_A - eta_AC G_B                                  : True
S Sinv = 1                                                          : True
S G_C S^-1 == sum_B L[B,C] G_B                                      : True
L^T eta L == eta                                                    : True
p on shell p.eta17.p == 1                                           : True
pslash^2 == 1                                                       : True
8 real solutions                                                    : True
classification                                                      : True
explicit quaternion commutant of G71R32                             : True
G17e[0]==[[0,1],[1,0]]x1, G17e[i]==diag(LO[i],-LO[i])               : True
Gamma17(x)^2 == (x.eta17.x) 1                                       : True
anticommutant of G17: dimension 1 (numerical rank 255 of 2048x256 system): True
G17[8] anticommutes with all G17[A], G17[8]^2 == -1                 : True
null k: k.eta17.k == 0, kslash^2 == 0, 8 rational solutions of kslash w = 0: True
Psi = w cos(k.x) is a REAL solution of G17^A d_A Psi = 0            : True
p.eta71.p == -m^2 and P32^2 == -m^2                                 : True
Psi32 (32 arbitrary real w) solves (G71R32^A d_A - m) Psi = 0       : True
Psi32 obeys the Klein-Gordon equation (box17 + m^2) Psi = 0         : True
(G71R32^A d_A)^2 == -box17 on 32 arbitrary functions                : True
```

4.4 Hashes of the files used to build 7+1.nb and its documentation

file	SHA-256
7+1.nb	f7abe7279b5540e914de876dc055c199bd491b9c2d9522bf28dbc23689e237ce
4+4.nb	740023233eaa0cd24ca0e2e7010094671b1d314b698114c08db94050d3e0aed5
tools/cayley_dickson_formulas.py	bdde8c4c8199b09ef36ae4e15d5d53bc62744fc85b4c5ba52eab5516f77b5d5f
tools/cayley_dickson_match.py	73f2cc584bbe1a3620c50890ffe5fb299857b04a14bc34fed2f9eaefd2e36c4e
tools/classify_clifford.py	42291276f46b046183fdbb7b62be7c5cedfd695fe9da96ac256afa23580930a9
tools/doc_common.py	8db6983bb3afce5ecdceba0b1ce95ed3ce5427d3b3acbd67651cfbd931e46e5b
tools/doc_provenance.py	12d4803e973f54837b9aa9aa92fb2bb0cc9a2342dc5cfad5439435174686d215
tools/doc_readme.py	211c31b72adab6d1da0cf8cd5880b221cf36446ab099f0eea2e5be0cd96aa64d
tools/doc_symbols.py	f7d7b363471d965d0e9fe900bebe6d10be700c3f02cb0684525fc9bbc5205b65
tools/docgen.py	bd229bca182d2de5939c5af0e9879e6ee22721629ee45518490b118661d76cfd
tools/extra71.py	9935ef99cd9300f2a7d4f06d36cf7e173912ebae61b6feff448ebe0588853168
tools/make_7p1_nb.py	6f617eb9d68adb69a8654529e5fb1d5287ee82e59ff48de96d373fcf6846c0ee
tools/make_docs.py	b89ef1f433cf259099cc359a0bdf9b2edb7d0fc148211890a953105a6877c05f
tools/nbparse.py	ad2255a8864be7447a55023840b327f789cb8c9d868d42261b19261653847423
tools/nbwrite.py	dcbe8ce5b3203c89a1d17fbe946e1b6b7bb7f8895afa944743d651cfefadeffb
tools/octalg.py	36b50120b69a963e252e07d60ec007755220822b4ad339716ea7aed93d58eb24
tools/ref44.py	ab100c3ebcc055fbe59c132a02f80ecb17630e934e99d59f88e919ec8064faf5
tools/ref71.py	4ac783b1ab221317387f2a74e0be60f114fd64940fe3ea96af79b33ee7dcb8b0
tools/run_eval.wl	f0f89ead42f002f839709b335be7007e0112c4d98f87a5768fd148bcac4de876
tools/verify_7p1_nb.py	0521c82e157d1712c48b51a572d3f8d28de58118aa7d639b4378fd39f31eb558

5. Process and environment

- Workspace: `D:\Developer\github\7+1` on Windows; Python and LaTeX ran in a Linux shell with that folder mounted. Intermediate files go to `build/`, which the repository's `.gitignore` excludes.
- The notebook is generated by `tools/make_7p1_nb.py` (cells written with `tools/nbwrite.py`), evaluated in place by `tools/run_eval.wl` in Wolfram 15.0.1, and cross-checked by `tools/verify_7p1_nb.py`.
- Python 3.10.12 with sympy 1.14.0 and numpy 2.2.6 (exact rational arithmetic for every identity).
- The documentation is generated by `tools/make_docs.py` from `tools/docgen.py`, `tools/doc_common.py`, `tools/doc_readme.py`, `tools/doc_symbols.py` and `tools/doc_provenance.py`: one source renders Markdown and LaTeX, and every matrix, table and check list is computed or parsed, never typed. The LaTeX is compiled with XeLaTeX (TeX Live 2022, `latexmk -xelatex`, fonts Latin Modern and DejaVu Sans Mono).
- History: the first algebraic experiments were made in a temporary cloud sandbox and then moved into `tools/` of the workspace, where everything was re-run. A Wolfram 15.0 kernel window was opened once by mistake before Wolfram 15.0.1 was available; no result comes from it. Only Wolfram 15.0.1 evaluated 7+1.nb.
- Corrections made during the review: an earlier draft called the real on-shell amplitudes of notebook section 5.4 "Majorana spinors"; that label was removed, and section 5.5 was added to show which real spinor FIELDS exist (massless with 16 components, massive only with the 32×32 matrices).
- The evaluated notebook was rendered section by section in the Wolfram 15.0.1 front end (`Rasterize`) and inspected. Characters that the front end's text font shows badly are written differently in text cells: the double-struck O and the arrow of $\vec{x}$ as small inline formulas, the script ℓ in the font Cambria Math, and $\not{p}$, $\not{k}$ as `pslash`, `kslash`.

6. Authorship

Notebook, tools and documentation written by Claude (Anthropic, model claude-opus-5-5) for Nash, on 2026-10-26, from the task statement and the notebook 4+4.nb described in section 1.